

LAB REPORT: Flood Inundation Mapping Using Sentinel-1 SAR **(Embakasi, Nairobi)**

1. Introduction

Flooding is one of the most frequent and destructive natural hazards affecting urban areas, particularly in rapidly growing cities such as Nairobi. The dense infrastructure and poor drainage systems in areas like Embakasi make them highly vulnerable to flood events.

This project aims to map flood inundation in Embakasi using **Sentinel-1 Synthetic Aperture Radar (SAR)** data. SAR is particularly useful for flood detection because it can penetrate cloud cover and operate both day and night. Flooded surfaces typically appear as **low backscatter (dark regions)** due to specular reflection of radar signals.

The workflow involves preprocessing SAR imagery, detecting changes, applying thresholds to extract flooded areas, refining the results, and computing flood statistics.

2. Study Area and Data

The study area is Embakasi Sub-County in Nairobi, Kenya. A polygon Region of Interest (ROI) was defined to spatially constrain the analysis.

Sentinel-1 Ground Range Detected (GRD) data was used with the following characteristics:

- **Sensor:** Sentinel-1 SAR
- **Polarization:** VH
- **Resolution:** 10 m
- **Mode:** Interferometric Wide Swath (IW)

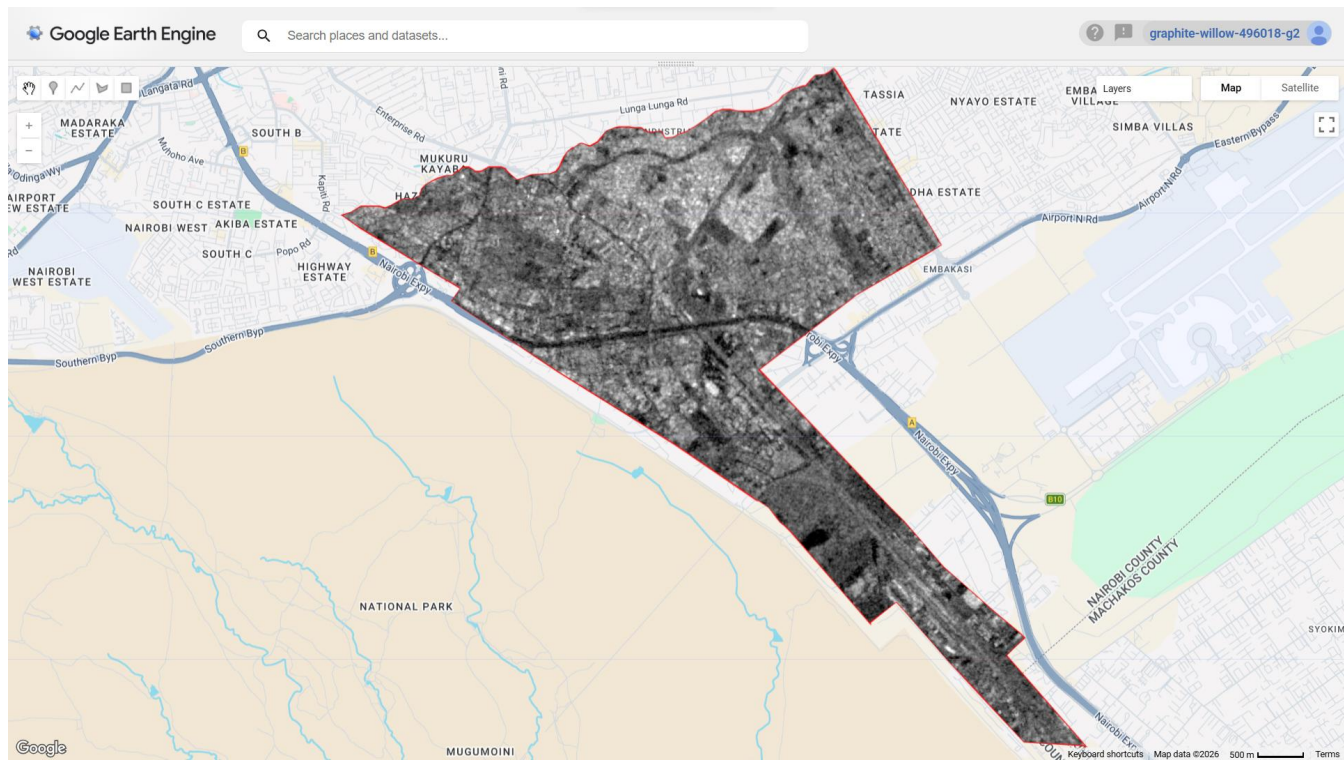
Two time periods were selected:

- **Before flood:** 2026-02-01 → 2026-02-12
- **After flood:** 2026-03-06 → 2026-03-18

Each period contained **3 scenes**, ensuring sufficient temporal coverage.

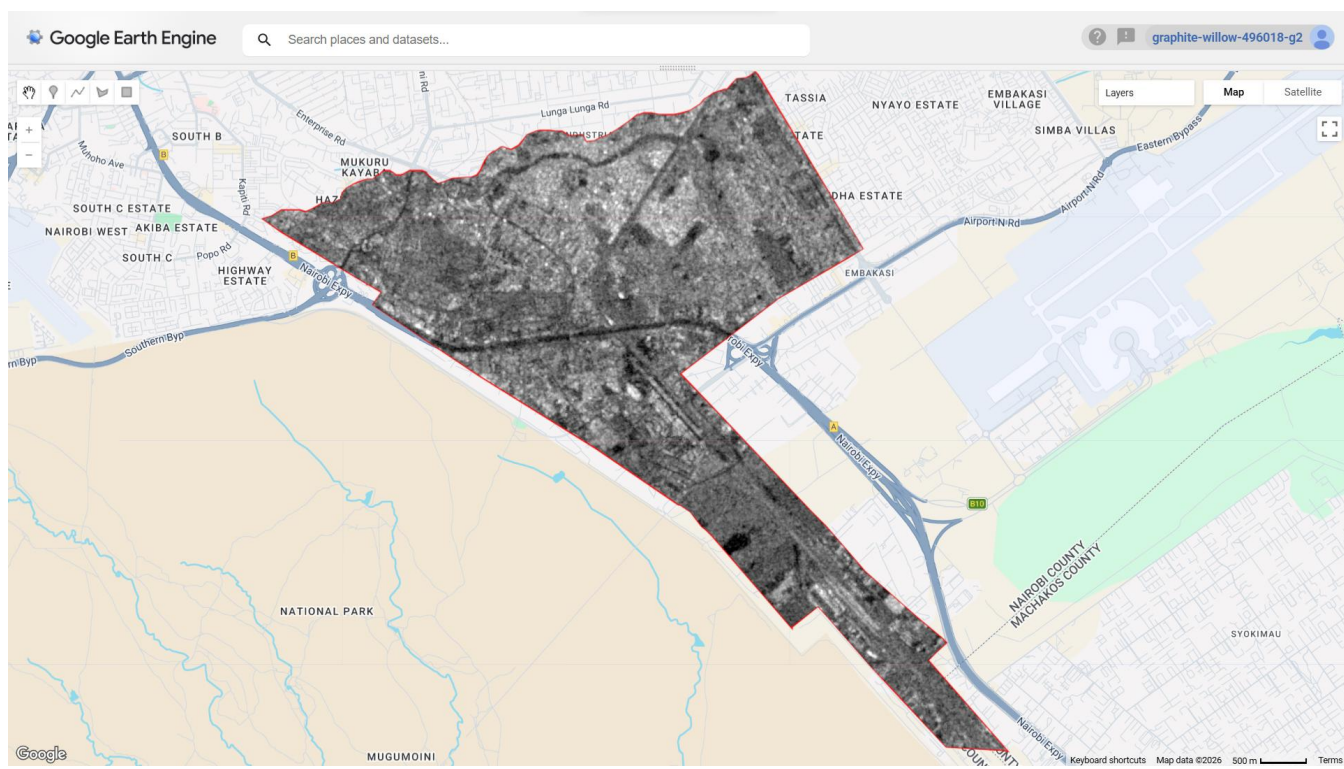
Screenshot 1: ROI boundary and raw Sentinel-1 images (Before vs After)

BEFORE



ROI boundary and raw Sentinel-1 image Before

AFTER



ROI boundary and raw Sentinel-1 image After

3. SAR Preprocessing (Step 3)

Speckle noise, inherent in SAR imagery, was reduced using a **focal mean filter** with a 50 m circular kernel.

This step improved image quality by:

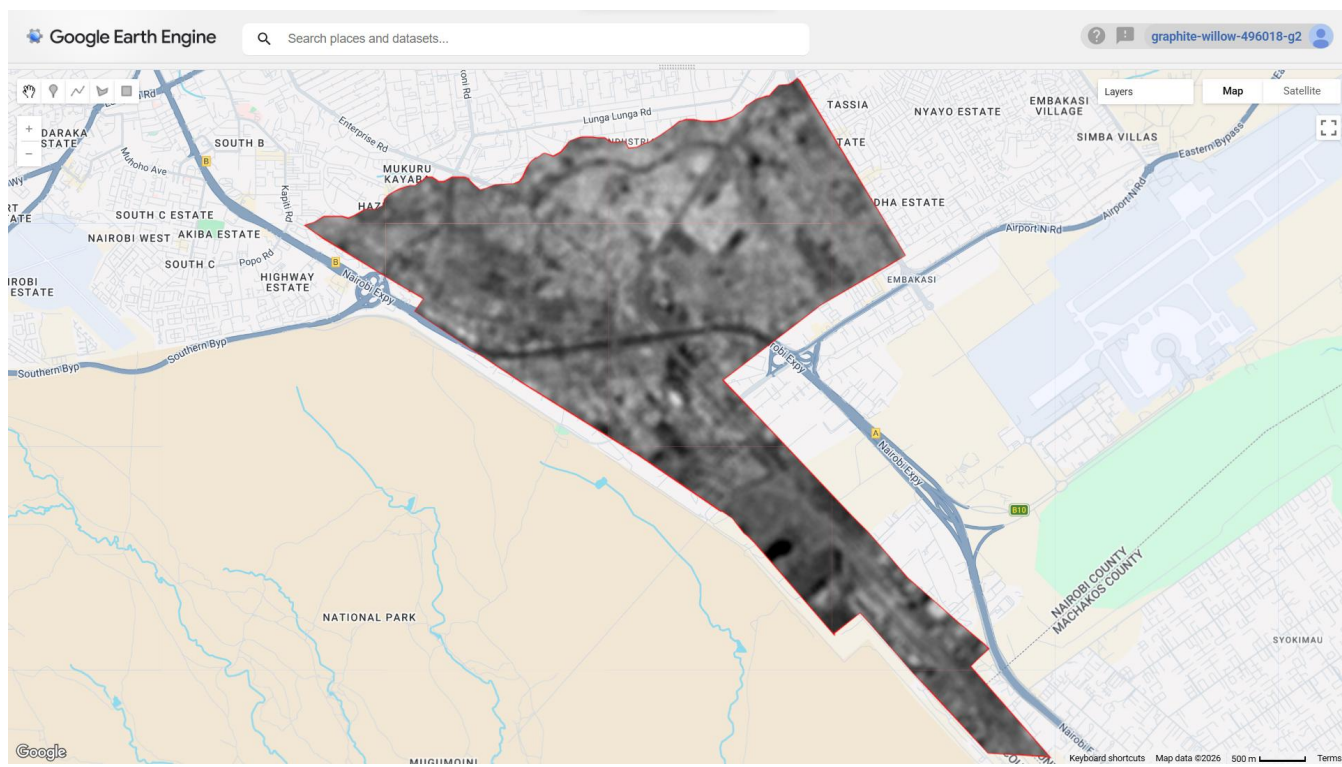
- Reducing salt-and-pepper noise
- Preserving general spatial patterns
- Enhancing flood detection reliability

Outputs:

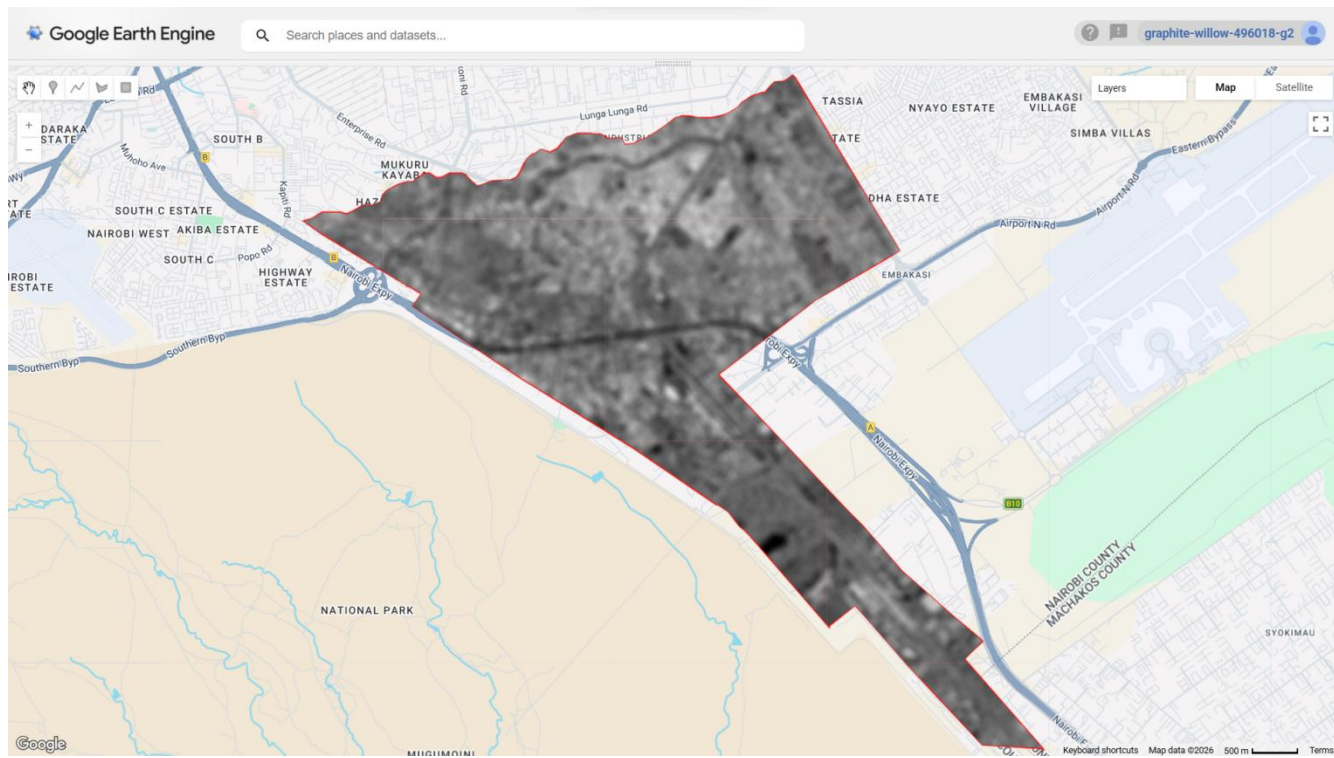
- beforeFiltered
- afterFiltered

Screenshot 2: Filtered SAR images (Before and After)

BEFORE



AFTER



4. Change Detection (Step 4)

Flood detection was performed using two complementary approaches:

4.1 Ratio Method

Ratio = After/ Before

Flooded areas show **low ratio values** (< 1).

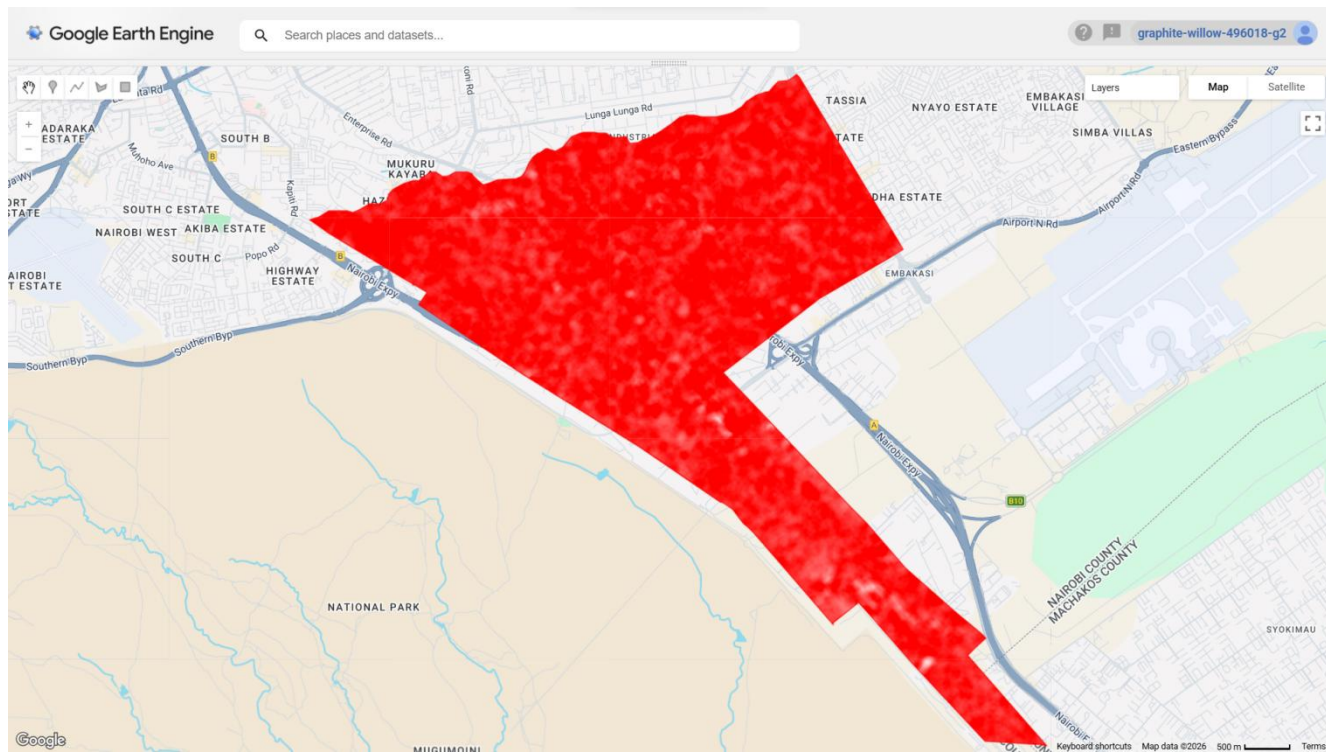
4.2 Difference Method

Difference = After–Before

Flooded areas show **negative values**.

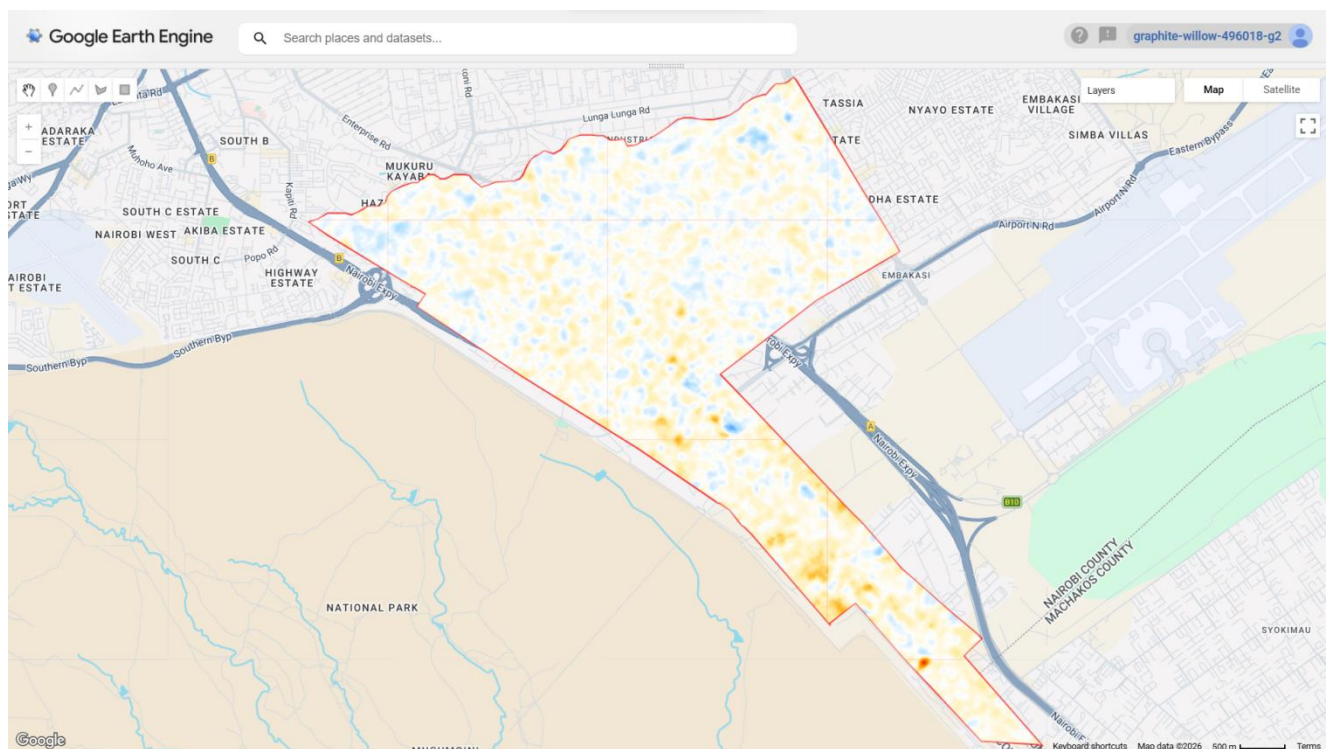
These methods highlight areas where radar backscatter decreased due to water presence.

RATIO IMAGE



Screenshot 3: Ratio image

DIFFERENCE IMAGE



Screenshot 4: Difference image

5. Flood Extraction (Step 5)

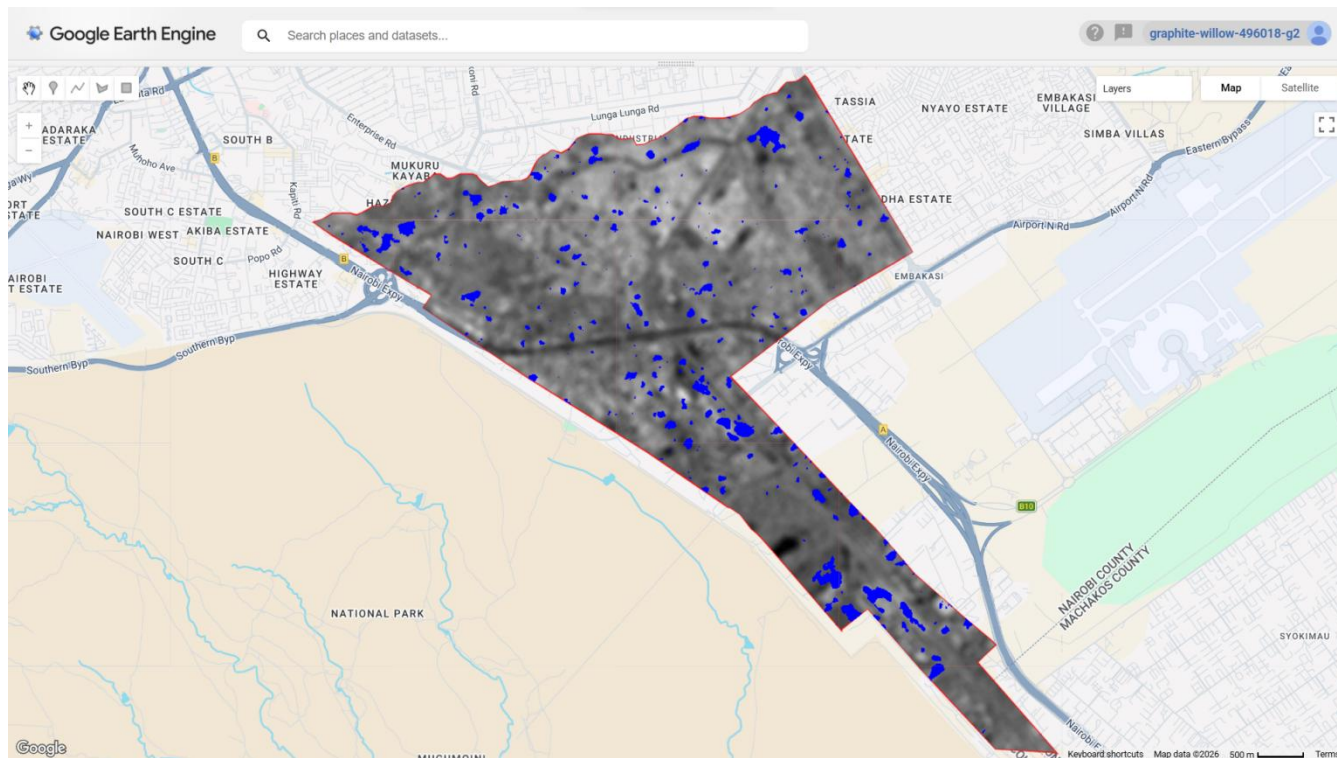
Initial thresholds were too strict and resulted in no detected flood. After debugging (Category D), thresholds were adjusted:

- **Ratio threshold:** 0.70 → **0.85**
- **Difference threshold:** -3 → **-1 dB**
- **Combination method:** AND → **OR**

The final flood mask (floodMaskFinal) produced:

- **Flooded area:** 0.8386 km²

FINAL FLOOD MASK



Screenshot 5: Final flood mask (Step 5 output)

6. Flood Mask Refinement (Step 6)

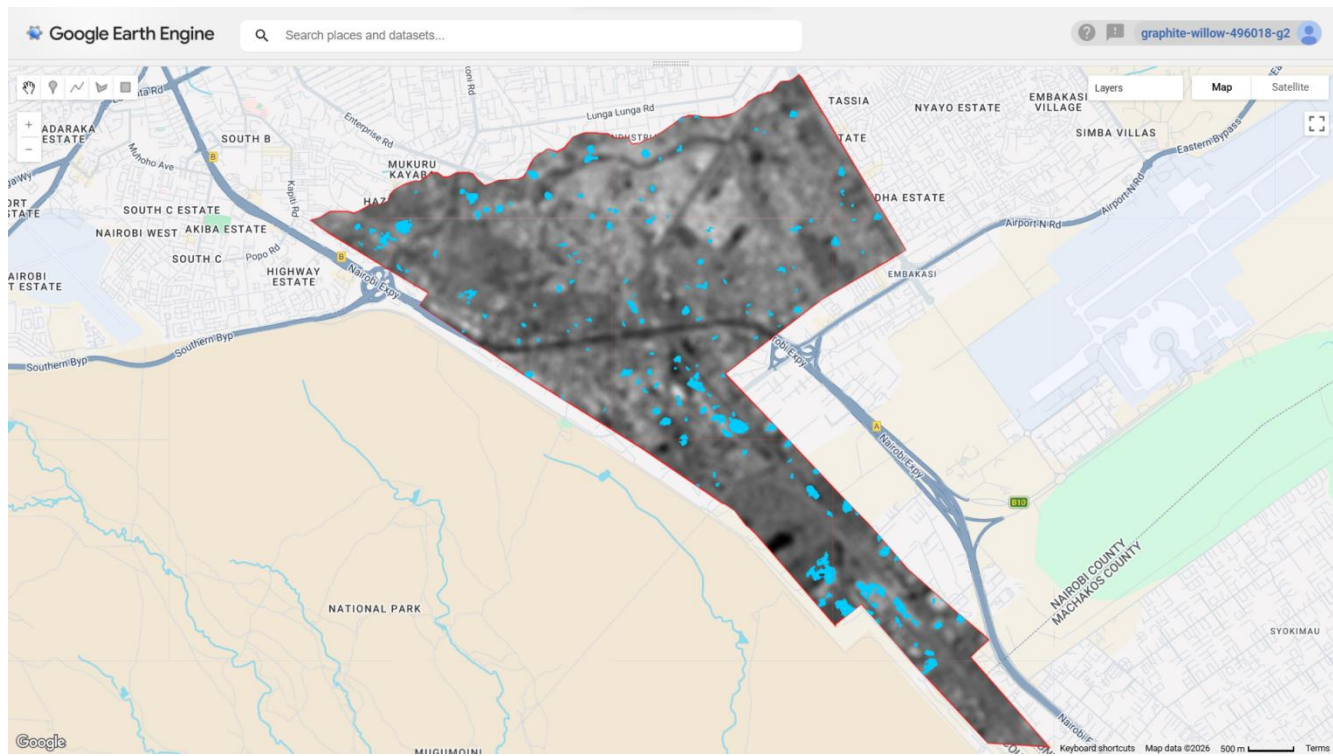
Three refinement processes were applied:

- Permanent water removal
- Slope masking (>5° removed)
- Morphological cleaning (min 8 pixels)

Results:

- Final flood area: **0.6595 km²**
- Retention: **78.64%**

FLOOD MASK REFINED

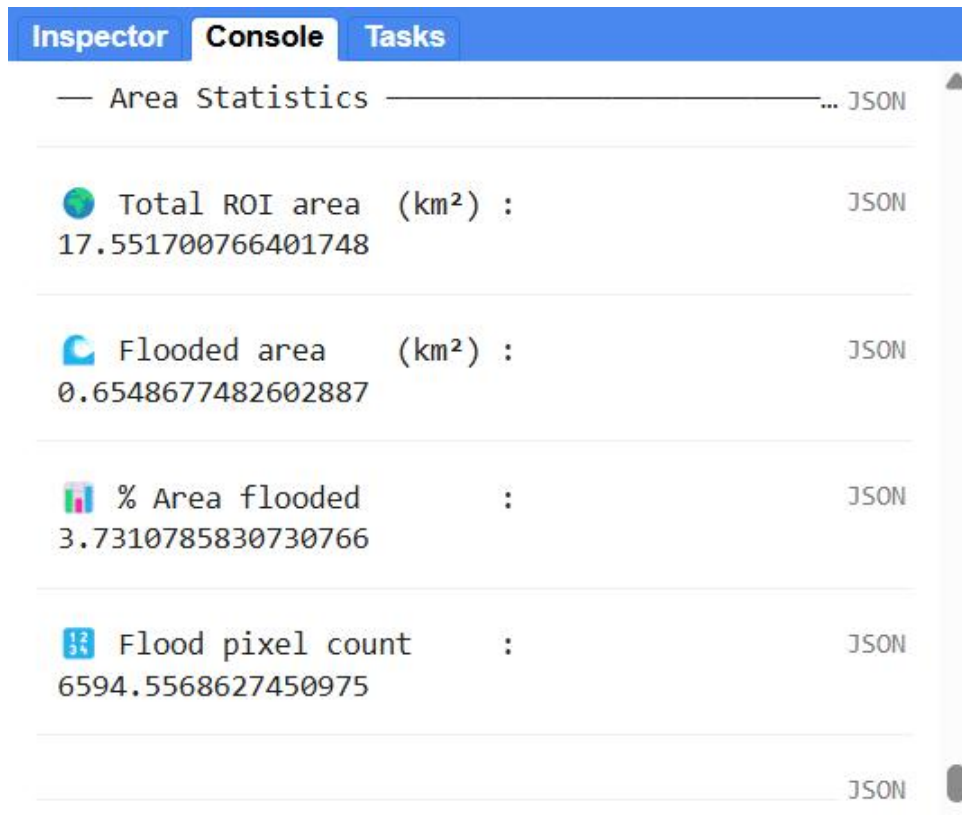


Screenshot 6: Refined flood mask (Step 6 output)

7. Final Statistics and Export (Step 7)

- **Total ROI area:** 17.55 km²
- **Flooded area:** 0.6549 km²
- **Percentage flooded:** 3.73%

FINAL STATISTICS CONSOLE OUTPUT

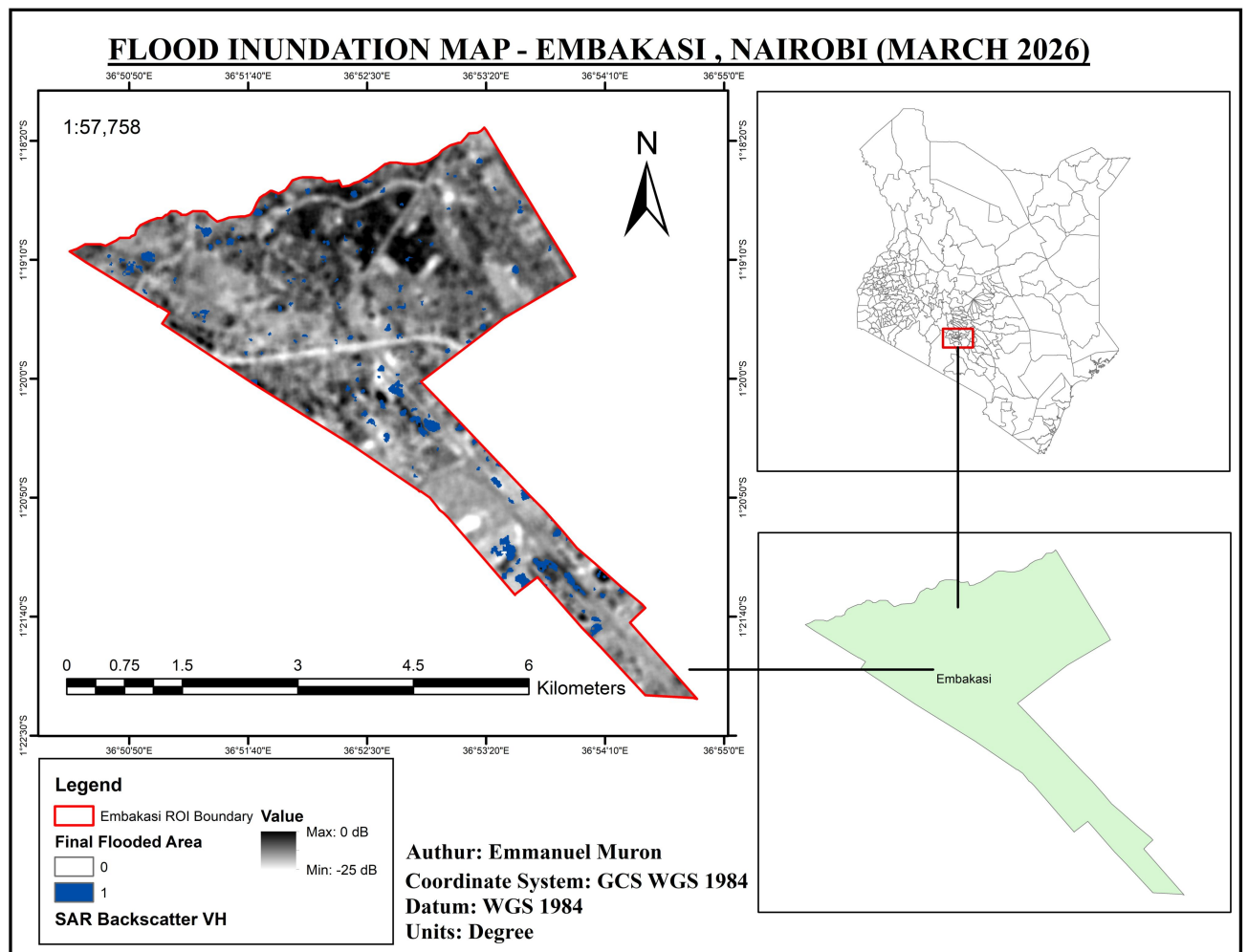


Screenshot 7: Final statistics console output

Table 1: Final Flood Statistics

Parameter	Value
Total ROI Area (km ²)	17.55
Flooded Area (km ²)	0.6549
Percentage Flooded (%)	3.73
Flood Pixel Count	6594.56

8. Final Map Presentation



FINAL MAP

9. Results and Interpretation

The flood mapping workflow successfully identified inundated areas within Embakasi Sub-County. The final refined flood extent covered approximately **0.6549 km²**, representing **3.73% of the total study area**.

The spatial distribution of flooded pixels corresponded well with expected low-lying regions and drainage pathways visible in the SAR imagery. Areas with reduced backscatter (dark tones) in the post-flood SAR image aligned with the extracted flood mask, confirming the effectiveness of the ratio and difference-based change detection methods.

The increase in detected flood area after threshold relaxation indicates that the initial thresholds were too conservative and failed to capture subtle backscatter variations. The use of the OR combination method ensured broader detection coverage, which is particularly important in emergency response scenarios.

Refinement steps further improved the reliability of the results by removing false positives associated with terrain slope and isolated noise pixels. The reduction from

0.8386 km² to 0.6595 km² demonstrates the importance of post-processing in flood mapping workflows.

10. Discussion

Despite the successful extraction of flood extent, several limitations were observed in the methodology.

Firstly, urban environments such as Embakasi present challenges due to **double-bounce scattering**, where radar signals reflect between buildings and water surfaces. This can cause flooded areas to appear bright instead of dark, leading to potential underestimation of flood extent.

Secondly, the reliance on threshold-based classification introduces subjectivity. The selected thresholds (ratio < 0.85 and difference < -1 dB) were determined empirically and may vary depending on environmental conditions, sensor geometry, and acquisition timing.

Additionally, the absence of permanent water removal effects in this case suggests that either the ROI contains minimal permanent water bodies or the threshold used (90%) was too strict to capture them.

Finally, SAR-based flood detection does not directly account for water depth or flow dynamics, limiting its ability to fully characterize flood severity.

Future improvements could include:

- Integration of optical imagery (e.g., Sentinel-2) where cloud-free data is available
 - Use of machine learning classification instead of fixed thresholds
 - Incorporation of elevation-based flood modeling
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11. Conclusion

This project successfully demonstrated flood mapping using Sentinel-1 SAR data in Embakasi. The use of change detection and adaptive thresholding enabled identification of flooded areas despite urban complexity.

Key insights:

- Threshold selection critically affects flood detection accuracy
- Urban environments introduce challenges such as double-bounce effects
- Refinement steps significantly improve reliability

The final estimated flooded area of **~0.65 km² (3.73%)** provides a realistic representation of flood extent and can support disaster response and planning.